

1. Firewall

Definition

A **Firewall** is a security system that monitors and controls incoming and outgoing network traffic based on predefined security rules.

Simple Idea

Think of a firewall as a **security guard at a building gate**:

- Allows authorized people to enter.
 - Blocks unauthorized people.
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Functions

- Blocks unauthorized access.
 - Allows legitimate traffic.
 - Monitors network activity.
 - Protects against attacks.
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Types

1. Hardware Firewall

- Physical device.
- Protects entire network.

2. Software Firewall

- Installed on a computer.
 - Protects a single device.
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Advantages

- Prevents unauthorized access.
 - Improves network security.
 - Filters malicious traffic.
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Exam Point

Firewall = First line of defense in network security.

2. Digital Signature

Definition

A **Digital Signature** is a cryptographic technique used to verify the authenticity and integrity of a message.

Provides

1. Authentication

Confirms sender identity.

2. Integrity

Ensures message was not modified.

3. Non-Repudiation

Sender cannot deny sending the message.

Working

Sender

1. Creates message.
2. Calculates hash of message.
3. Encrypts hash using private key.
4. Result is Digital Signature.

Receiver

1. Receives message and signature.
2. Decrypts signature using sender's public key.
3. Computes hash of received message.
4. Compares hashes.

If hashes match:

- Signature is valid.
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Exam Point

Digital Signature uses Private Key for signing and Public Key for verification.

3. Authentication Mechanisms and Protocols

Authentication

Definition

Authentication is the process of verifying the identity of a user or system.

Authentication Factors

1. Something You Know

- Password
- PIN

2. Something You Have

- Smart card
- Token

3. Something You Are

- Fingerprint
 - Face recognition
 - Iris scan
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Authentication Protocol

Definition

A set of rules used to verify identity securely over a network.

Common Protocols

PAP

Password Authentication Protocol

- Sends password directly.
- Less secure.

CHAP

Challenge Handshake Authentication Protocol

- Uses challenge-response mechanism.
- More secure.

Kerberos

- Uses tickets.
 - Provides secure authentication.
 - Widely used in organizations.
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Exam Point

Authentication answers:

"Who are you?"

Authorization answers:

"What are you allowed to do?"

4. RSA (Public Key Cryptography)

Definition

RSA is an asymmetric encryption algorithm that uses two keys:

Public Key

Used for encryption.

Private Key

Used for decryption.

Why RSA?

Solves the problem of securely sharing keys.

Working

Encryption

Sender encrypts message using:

[
$$\text{Ciphertext} = \text{Message}^{\{e\}} \bmod n$$

]

using Public Key.

Decryption

Receiver decrypts using:

[
$$\text{Message} = \text{Ciphertext}^{\{d\}} \bmod n$$

]

using Private Key.

Uses

- Secure communication.
 - Key exchange.
 - Digital signatures.
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Advantages

- No need to share private key.
 - High security.
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Exam Point

Public Key

Can be shared with everyone.

Private Key

Must remain secret.

5. AES (Advanced Encryption Standard)

Definition

AES is a symmetric encryption algorithm used worldwide for protecting data.

Symmetric Encryption

Same key is used for:

- Encryption
- Decryption

Key Sizes

AES-128

128-bit key

AES-192

192-bit key

AES-256

256-bit key

More bits = More security

Working Stages

1. SubBytes

Byte substitution.

2. ShiftRows

Rows shifted.

3. MixColumns

Columns mixed.

4. AddRoundKey

Round key added.

Uses

- Wi-Fi security
- VPNs
- Banking systems

- File encryption
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Advantages

- Fast
 - Secure
 - Widely used
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Exam Point

AES = Symmetric Encryption

RSA = Asymmetric Encryption

Most Important Differences

Feature	AES	RSA
Type	Symmetric	Asymmetric
Keys	One Key	Two Keys
Speed	Fast	Slower
Use	Data Encryption	Key Exchange, Signatures
Security	Strong	Strong

2-Minute Revision Before Exam

Firewall

- Filters network traffic.
- Blocks unauthorized access.
- First line of defense.

Digital Signature

- Authentication.
- Integrity.
- Non-repudiation.

- Sign with Private Key.
- Verify with Public Key.

Authentication

- Verifies identity.
- Factors: Know, Have, Are.
- Protocols: PAP, CHAP, Kerberos.

RSA

- Asymmetric encryption.
- Public Key encrypts.
- Private Key decrypts.
- Used in digital signatures.

AES

- Symmetric encryption.
- Same key for encryption and decryption.
- Fast and highly secure.
- AES-128, AES-192, AES-256.

Golden Exam Question

Difference between RSA and AES

- RSA uses **two keys (public/private)** and is mainly used for **key exchange and digital signatures**.
- AES uses **one shared key** and is mainly used for **fast data encryption**.